Introduction

Project title: **AI STUDYBUDDY**

Team ID : SWTID-2026-8810

Team Size : 4

Team Leader : Kalaiyarasi S

Team member : Thasmiya G

Team member : Suba Sree M

Team member : Mahalakshmi V

1.Project Overview

AI Study Buddy is an educational application developed to help students improve their learning experience with the support of Artificial Intelligence. The system allows students to ask academic questions, receive simple explanations, generate study notes and quizzes, and organize their learning activities.

It provides a user-friendly platform where students can access personalized study assistance based on their needs.

The project uses HTML, CSS, and JavaScript for the frontend, Node.js and Express.js for the backend, MongoDB for storing user and study data, and the Gemini API for AI-based responses.

Postman is used to test the APIs, and JWT can be used for secure user authentication. The main objective of AI Study Buddy is to make studying easier, more interactive, and more effective for students.

AI Study Buddy acts as a virtual study companion that helps students understand difficult topics, ask questions, generate study materials, and practice through quizzes. It uses Artificial Intelligence to make the learning process more interactive and personalized. AI Study Buddy aims to save students' time, improve their understanding of subjects, and encourage independent learning.

2.Project Objectives

The project aims to make learning more interactive, accessible, and effective by using Artificial Intelligence.

OBJECTIVES

- To provide students with a smart and easy-to-use study assistant.
- To answer academic questions quickly and clearly.
- To explain difficult topics in a simple way.
- To generate study notes and learning materials.
- To create quizzes for practice and self-assessment.
- To support personalized learning based on student needs.
- To help students organize their study activities.
- To make learning more interactive and effective.
- To save students' time during study and revision.

3.Target users

- **School Students** – For learning subjects, clearing doubts, and practicing quizzes.
- **College Students** – For academic preparation, notes, and revision.
- **Competitive Exam Students** – For practice questions and quick topic explanations.
- **Teachers** – For preparing study materials and quiz questions.
- **Self-Learners** – For learning new topics independently.
- **Students Preparing for Exams** – For revision, practice, and study planning

4.Pre-requisite & knowledge requirements

1. **Programming Language & Runtime**
   JavaScript (ES6+), basic Node.js concepts, functions, arrays, objects, and asynchronous programming. Basic knowledge of npm and package management.
2. **Frameworks & Development Tools**
   Node.js and Express.js for backend development, Visual Studio Code for coding, and Postman for testing API endpoints.
3. **Database Management (MongoDB)**
   Basic knowledge of MongoDB, collections, documents, CRUD operations, and connecting MongoDB with Node.js.
4. **AI & API Integration**
   Basic understanding of Gemini API, API keys, sending requests, receiving responses, and integrating AI features into the application.
5. **Authentication & Security**
   Basic knowledge of user registration and login, JWT authentication, password security, and secure API access.

5.Project architecture and tech stack

1. Project Architecture

Frontend → Backend API → AI Service → Database

Frontend Layer: Provides the user interface for registration, login, asking questions, study notes, and quizzes.
Backend Layer: Node.js and Express.js handle user requests, business logic, and API communication.
AI Layer: Gemini API processes student questions and generates study-related responses.

Database Layer: MongoDB stores user details, study data, questions, and responses.
Authentication Layer: JWT is used to manage secure user authentication.

2. Technology Stack

Component	**Technology**
Frontend	HTML, CSS, JavaScript
Backend	Node.js, Express.js
Database	MongoDB
AI Service	Gemini API
API Testing	Postman
Authentication	JWT
Code Editor	Visual Studio Code
Data Format	JSON

6.Project setup and configuration

1. Create Project Folder
Create a separate folder for the AI Study Buddy project and open it in Visual Studio Code.
2. Initialize Node.js Project
Initialize the project using npm and create the package.json file.
3. Install Required Packages
Install Express.js, MongoDB/Mongoose, CORS, dotenv, JWT, and other required dependencies.
4. Configure Backend
Create the Node.js and Express.js server and configure the required API routes.
5. Configure MongoDB
Connect the application to MongoDB and create the required collections for storing user and study data.
6. Configure Gemini API

Add the Gemini API key using environment variables and configure the AI service for generating study responses.

7. Configure Authentication
Set up user registration, login, and JWT-based authentication.
8. Test APIs
Use Postman to test registration, login, AI study, and other API endpoints.

7. Features & api endpoints

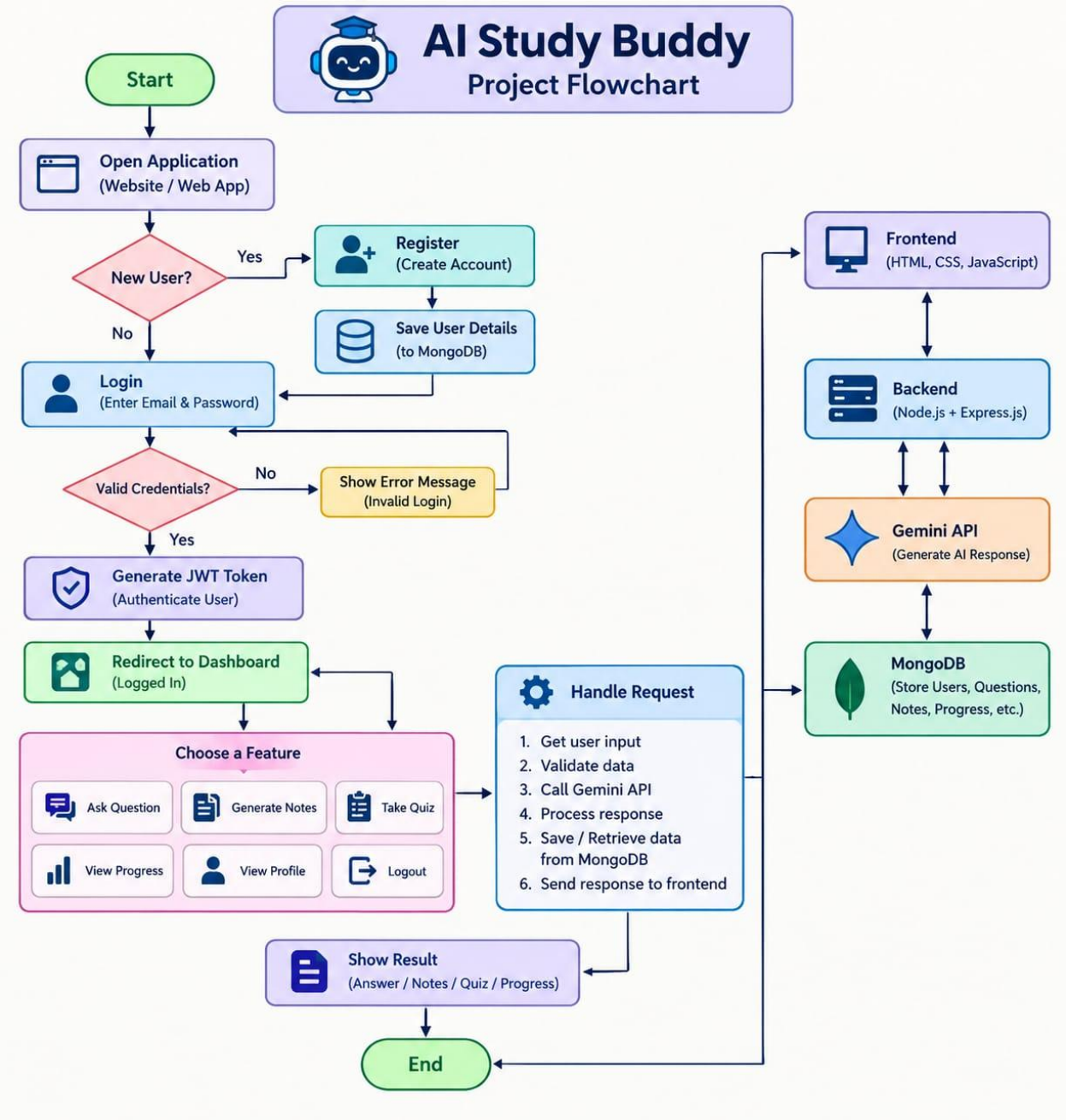
Features

- **User Registration** – Allows new users to create an account.
- **User Login** – Provides secure access to registered users.
- **AI Study Assistant** – Answers students' academic questions using Gemini API.
- **Study Notes Generation** – Helps generate simple and useful study notes.
- **Quiz Generation** – Creates practice questions for students.
- **Personalized Learning** – Provides study support based on the user's requirements.
- **Progress Tracking** – Helps users monitor their study activities.
- **Secure Authentication** – Uses JWT for secure user authentication.

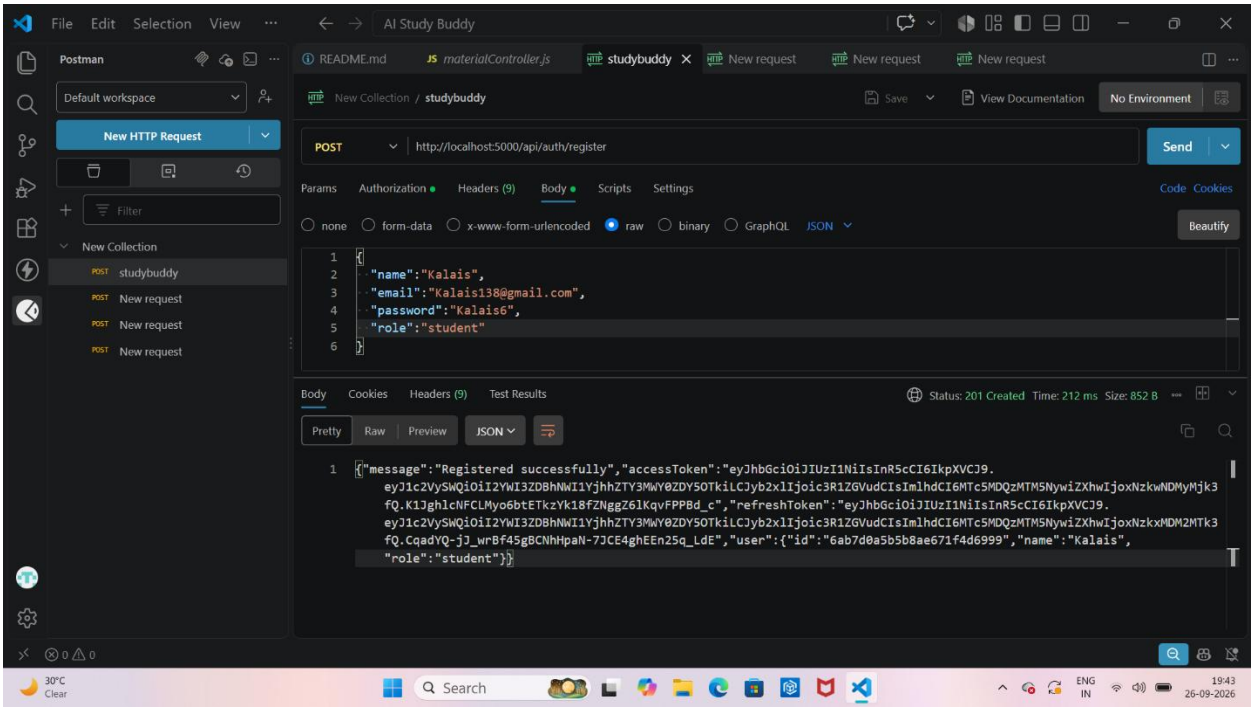
API Endpoints

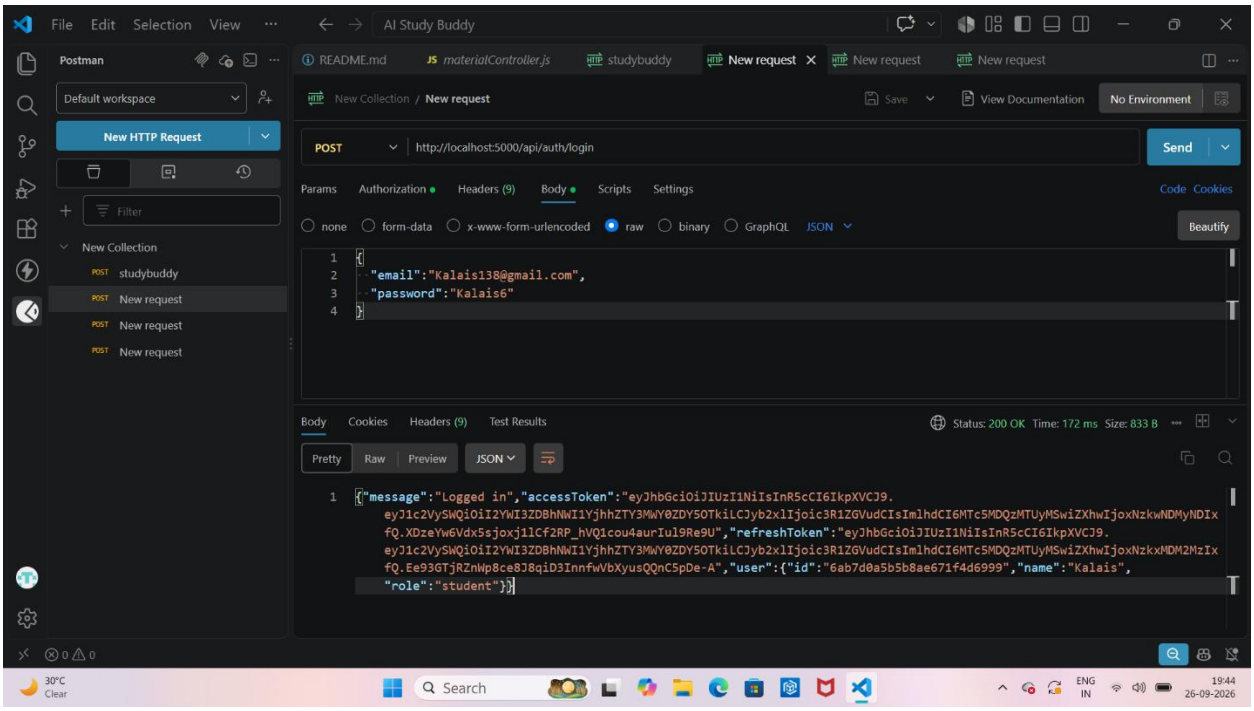
Method	Endpoint	Purpose
POST	/api/register	Register a new user
POST	/api/login	Login user
POST	/api/ask	Ask an academic question
POST	/api/notes	Generate study notes
POST	/api/quiz	Generate quiz questions
GET	/api/progress	View study progress
GET	/api/profile	View user profile
POST	/api/logout	Logout user

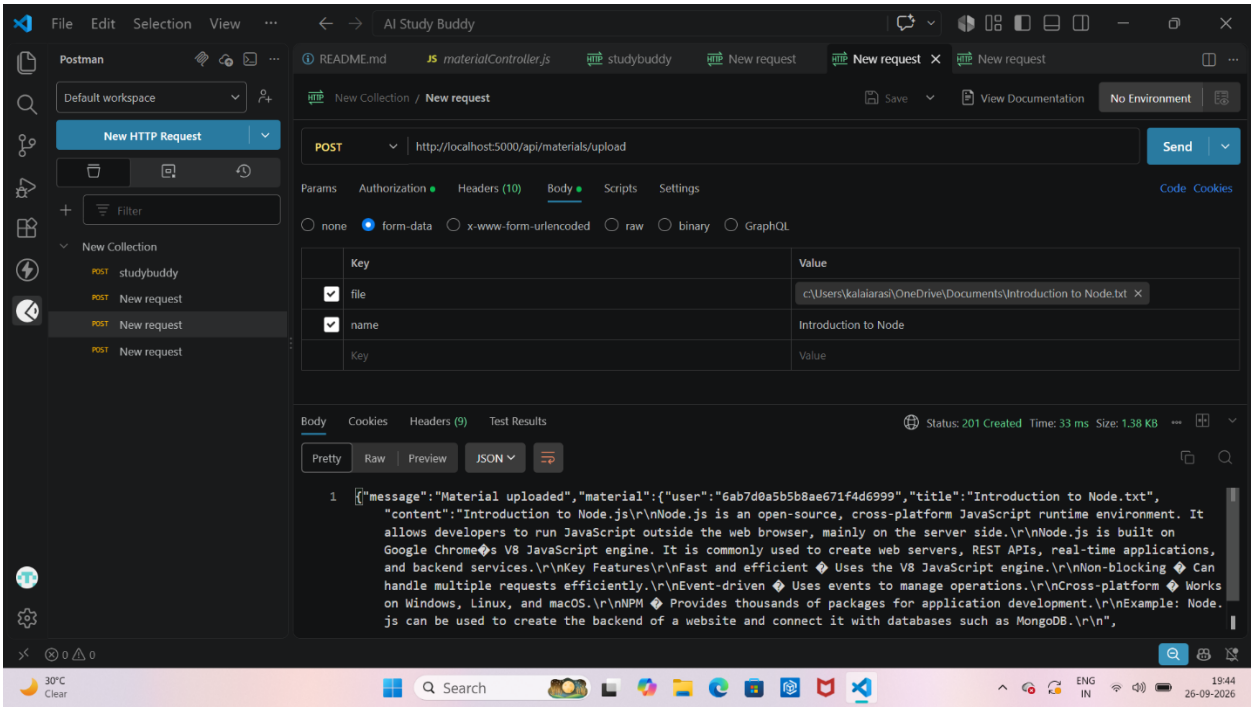
8.Project flowchart

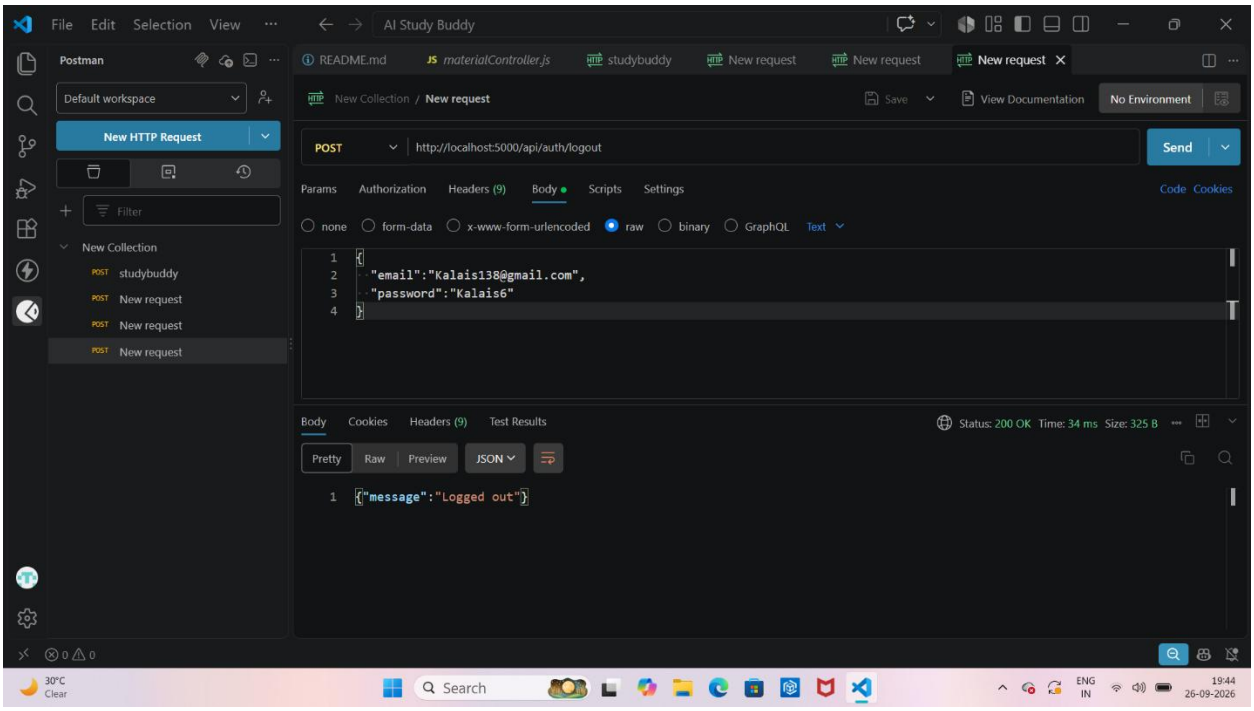


9. Output


File Edit Selection View
AI Study Buddy
Postman
Default workspace
New HTTP Request
New Collection
studybuddy
New request
New request
New request
README.md
materialController.js
studybuddy
New request
New request
New request
New Collection / studybuddy
Save
View Documentation
No Environment
POST
http://localhost:5000/api/auth/register
Send
Params
Authorization
Headers (9)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"name":"Kalais",
"email":"Kalais138@gmail.com",
"password":"Kalais6",
"role":"student"
}
Body
Cookies
Headers (9)
Test Results
Status: 201 Created
Time: 212 ms
Size: 852 B
Pretty
Raw
Preview
JSON
{"message":"Registered successfully","accessToken":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJ1c2VySWQiOiI2YWI3ZDBhNWI1YjhhZTY3MWY0ZDY5OTkiLCJyb2xlIjoic3R1ZGVudCIsImlhdCI6MTc5MDQzMTM5NywiZXhwIjoxNzkwNDMyMjk3
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eyJ1c2VySWQiOiI2YWI3ZDBhNWI1YjhhZTY3MWY0ZDY5OTkiLCJyb2xlIjoic3R1ZGVudCIsImlhdCI6MTc5MDQzMTM5NywiZXhwIjoxNzkxMDM2MTk3
fQ.CqadYQ-jJ_wrBf45gBCNhHpaN-7JCE4ghEEn25q_LdE","user":{"id":"6ab7d0a5b5b8ae671f4d6999","name":"Kalais",
"role":"student"}}
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Search
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26-09-2026


File Edit Selection View
AI Study Buddy
Postman
Default workspace
New HTTP Request
New Collection
studybuddy
New request
New request
New request
README.md
materialController.js
studybuddy
New request
New request
New request
New Collection / New request
Save
View Documentation
No Environment
POST
http://localhost:5000/api/auth/login
Send
Params
Authorization
Headers (9)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Beautify
{
"email":"Kalais138@gmail.com",
"password":"Kalais6"
}
Body
Cookies
Headers (9)
Test Results
Status: 200 OK
Time: 172 ms
Size: 833 B
Pretty
Raw
Preview
JSON
{"message":"Logged in","accessToken":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJ1c2VySWQiOiI2YWI3ZDBhNWI1YjhhZTY3MWY0ZDY5OTkiLCJyb2xlIjoic3R1ZGVudCIsImlhdCI6MTc5MDQzMTUyMSwiZXhwIjoxNzkwNDMyNDIx
fQ.XDzeYw6Vdx5sjoxj1lCf2RP_hVQ1cou4aurIul9Re9U","refreshToken":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
eyJ1c2VySWQiOiI2YWI3ZDBhNWI1YjhhZTY3MWY0ZDY5OTkiLCJyb2xlIjoic3R1ZGVudCIsImlhdCI6MTc5MDQzMTUyMSwiZXhwIjoxNzkxMDM2MzIx
fQ.Ee93GTjRZnWp8ce8J8qiD3InnfwVbXyusQQnC5pDe-A","user":{"id":"6ab7d0a5b5b8ae671f4d6999","name":"Kalais",
"role":"student"}}
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26-09-2026


File Edit Selection View
AI Study Buddy
Postman
Default workspace
New HTTP Request
New Collection
studybuddy
New request
New request
New request
README.md
materialController.js
studybuddy
New request
New request
New request
New Collection / New request
Save
View Documentation
No Environment
POST
http://localhost:5000/api/materials/upload
Send
Params
Authorization
Headers (10)
Body
Scripts
Settings
Code
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
file
c:\Users\kalaiaras\OneDrive\Documents\Introduction to Node.txt
name
Introduction to Node
Key
Value
Body
Cookies
Headers (9)
Test Results
Status: 201 Created
Time: 33 ms
Size: 1.38 KB
Pretty
Raw
Preview
JSON
{"message":"Material uploaded","material":{"user":"6ab7d0a5b5b8ae671f4d6999","title":"Introduction to Node.txt",
"content":"Introduction to Node.js\r\nNode.js is an open-source, cross-platform JavaScript runtime environment. It
allows developers to run JavaScript outside the web browser, mainly on the server side.\r\nNode.js is built on
Google Chrome's V8 JavaScript engine. It is commonly used to create web servers, REST APIs, real-time applications,
and backend services.\r\nKey Features\r\nFast and efficient • Uses the V8 JavaScript engine.\r\nNon-blocking • Can
handle multiple requests efficiently.\r\nEvent-driven • Uses events to manage operations.\r\nCross-platform • Works
on Windows, Linux, and macOS.\r\nNPM • Provides thousands of packages for application development.\r\nExample: Node.
js can be used to create the backend of a website and connect it with databases such as MongoDB.\r\n",
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10.Conclusion

AI Study Buddy is a user-friendly educational application that provides students with a convenient platform for learning and academic support. It helps users clear their doubts, understand difficult topics, prepare study notes, and practice through quizzes.

The integration of the Gemini API provides AI-based responses, while Node.js, Express.js, and MongoDB support the backend and data management. The system also provides secure user authentication and API-based communication.

Overall, AI Study Buddy can make the learning process more interactive, personalized, organized, and effective, while encouraging students to learn independently and improve their academic performance.